

LR(0) - Results



Attempt 1 of 1

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Attempt Score **10 / 10 - 100 %**

Overall Grade (Highest Attempt) **10 / 10 - 100 %**

Question 1

1 / 1 point

The accept action happens when

- ☐ The stack is empty
- ☐ The input is empty regardless of stack
- ☒ ACTION[state, \$] = accept
- ☐ GOTO[state, S] = accept

Question 2

1 / 1 point

A shift/reduce conflict occurs when

- ☐ Two reductions are possible
- ☒ Both shift and reduce are possible on same (state, lookahead)
- ☐ No action is defined

- ☐ GOTO leads to multiple states

Question 3

1 / 1 point

In an item like $A \rightarrow \alpha \cdot \beta$, the dot indicates

- ☐ The end of input
- ✓ ☒ How many symbols of the RHS have been seen
- ☐ The lookahead position
- ☐ The next state number

Question 4

1 / 1 point

The closure operation on a set of items mainly adds

- ✓ ☒ Items for productions whose LHS appears after the dot
- ☐ Only completed items
- ☐ Only items with \$ lookahead
- ☐ Items for terminals only

Question 5

1 / 1 point

The next four questions are based on the following grammar:

- (0) $E' \rightarrow E$
- (1) $E \rightarrow E + T$
- (2) $E \rightarrow T$
- (3) $T \rightarrow id$

What is GOTO(I0,id)?

- ✓ ☒ { $T \rightarrow id \bullet$ }
- ☐ { $T \rightarrow id \bullet, E \rightarrow T \bullet$ }
- ☐ { $E \rightarrow id \bullet$ }
- ☐ { $E \rightarrow T \bullet, T \rightarrow id \bullet, E' \rightarrow E \bullet$ }

Question 6

1 / 1 point

Suppose a state contains the item $E \rightarrow E \bullet + T$. What is GOTO(thatState, +)?

- ☐ { $E \rightarrow E + \bullet T$ }
- ✓ ☒ { $E \rightarrow E + \bullet T, T \rightarrow \bullet id$ }
- ☐ { $T \rightarrow id \bullet$ }
- ☐ { $E \rightarrow \bullet T, T \rightarrow \bullet id$ }

Question 7

1 / 1 point

Which state is a reduce state?

- ☐ { $T \rightarrow \bullet id$ }
- ☐ { $E' \rightarrow E \bullet, E \rightarrow E \bullet + T$ }
- ✓ ☒ { $E \rightarrow T \bullet$ }
- ☐ { $E \rightarrow E + \bullet T, T \rightarrow \bullet id$ }

☐ more than one answer

Question 8

1 / 1 point

Which item indicates the **accept** configuration?

- ☐ $E \rightarrow E + T \bullet$
- ✓ ☒ $E' \rightarrow E \bullet$
- ☐ $E \rightarrow T \bullet$
- ☐ $T \rightarrow id \bullet$

Question 9

1 / 1 point

A shift-reduce conflict in LR(0) happens when:

- ☐ A state has two completed items
- ✓ ☒ A state has both a completed item and a shift possibility
- ☐ A state has two shift entries
- ☐ Grammar is left recursive

Question 10

1 / 1 point

In LR(0), a reduce action is entered:

- ☐ For specific lookahead terminals
- ☐ For all terminals in FOLLOW
- ✓ ☒ For all terminals in grammar

☐ Only for \$

Done